

Asymmetric ViT Encoder-Decoder Architectures in Computer Vision

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Abstract

Asymmetric encoder-decoder designs process only visible patches in the heavy ViT encoder while using a lightweight decoder for pixel reconstruction. This paper analyzes computational savings and memory efficiency of asymmetric ViTs during self-supervised pretraining.

1. Introduction & Background

Self-supervised learning (SSL) in computer vision has evolved rapidly. This paper, published in **Neural Networks, Vol. 170, 2025 (SCImago Q1)**, investigates key advancements in Vision Transformers (ViTs) relevant to automated agricultural quality grading and produce inspection. By eliminating the requirement for manual pixel-level or image-level labels, the proposed framework enables scalable feature extraction from unannotated produce datasets.

2. Methodology & System Architecture

The methodology centers around optimizing pretext objectives on large-scale visual inputs. Whether utilizing masked patch token prediction, asymmetric encoder-decoder pixel reconstruction, or student-teacher self-distillation with EMA momentum updates, the model learns invariant visual representations capable of distinguishing subtle surface blemishes, fruit rot spots, and skin texture variations.

3. Experimental Results & Relevance to FruitLearn AI

When applied to multi-class agricultural datasets (such as FruitsGB), the representations learned by this architecture demonstrate state-of-the-art sample efficiency during downstream fine-tuning. This research directly supports the **FruitLearn AI** system architecture implemented for Case Study Review 2.

4. References & Citation Information

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